

**FORECASTING COVID-19 WITH
TEMPORAL HIERARCHIES AND ENSEMBLE METHODS**

A Thesis Presented

by

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Biostatistics and Epidemiology

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ABSTRACT

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Infectious disease forecasting efforts underwent rapid growth during the COVID-19 pandemic, providing guidance for pandemic response and about potential future trends. Yet despite their importance, short-term forecasting models often struggled to produce accurate real-time predictions of a complex and rapidly changing system. This gap in accuracy, which persists three years into the pandemic, warrants the exploration and testing of new methods to glean fresh insights.

In this work, we examine the application of the temporal hierarchical forecasting (THieF) methodology to probabilistic forecasts of COVID-19 incident hospital admissions in the United States. THieF is a recent, innovative forecasting technique that aggregates time-series data into a hierarchy made up of different temporal scales, produces forecasts at each level of the hierarchy, then reconciles those forecasts using optimized weighted forecast combination. While THieF's unique approach has shown substantial accuracy improvements in a diverse range of applications, such as opera-

tions management and emergency room admission predictions, this technique has not previously been applied to outbreak forecasting.

We generate candidate models formulated using the THieF methodology, which differ by their hierarchy schemes and data transformations, and weighted ensembles of the THieF models. The models are validated using weighted interval score (WIS), and the top-performing subset is compared to several benchmark models. These models include simple SARIMA models, a naive baseline model, and a median ensemble of operational incident hospitalization models from the US COVID-19 Forecast Hub. The THieF models and THieF ensembles demonstrate improvements in WIS and MAE, as well as competitive prediction interval coverage, over the benchmark models for both the validation and testing phases. These accuracy improvements suggest the THieF methodology may serve as a promising new technique for infectious disease forecasting that makes better predictions in real-time.

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CHAPTER 1

INTRODUCTION

- 1.1 Infectious disease forecasting**
- 1.2 COVID-19 forecasting**
 - 1.2.1 US COVID-19 Forecast Hub**
- 1.3 Temporal hierarchical forecasting**
- 1.4 Goals**

CHAPTER 2

METHODS

2.1 Surveillance data

- data used: HealthData.gov COVID-19 Reported Patient Impact and Hospital Capacity by State Time Series (reports daily incident hospitalizations for all 50 states, several US territories, and the AGGREGATED US national level) - dataset sources: HHSTeleTracking, state/territorial health departments which report to HHS Protect, and National Health Care Safety Network (prior to July 15, 2020). Note that the first date of truth data used is July 27, 2020 (see Date range of analysis) - This data set designated official ground truth data for COVID-19 incident hospitalizations on December 1, 2020 by the US COVID-19 Forecast Hub [<https://covid19forecast-hub.org/>], for usage as both input training data and to compare against for accuracy evaluations - hospitalization reporting sporadic for first four months of pandemic, consistency only achieved for all locations in late July 2020 (all territories designated "states" had consistent reporting starting [date]) - Hospitalization data became more reliable compared to case and death data due to reporting for Medicare reimbursement. Case and death data have suffering major under reporting issues that only grew worse as the pandemic continued on - Truth data collection for COVID-19 incident cases and incident deaths was discontinued in March 2023, with John Hopkins University (the official ground truth data source for those targets) shutting down their COVID dashboard(?) - Note though that hospitalization data is not perfect and values may be updated, especially within the first few weeks and incident hospitalization value is posted. However, models are training on the currently available data to them,

which may not reflect the finalized truth value - Truth data is queried through Zoltar, a data repository set up by..., which includes past versions of the truth data so that original values may be obtained. The data used to train the models in this analysis is queried such that it would be the data available as of the date that the retrospective forecast is "made on" so as to be comparable to other models.

2.2 Forecasts

2.2.1 Forecast quantiles

2.2.2 Forecast locations

2.2.3 Forecast horizons

2.3 Date range of analysis

2.4 Comparison models

- COVIDhub-baseline: A naive baseline model created and run by the US COVID-19 Forecast Hub meant to serve as a benchmark model to compare against. This model makes forecasts using the previous day's incident hospitalization value as the point forecast (and 0.5 quantile for the probabilistic forecast) and past changes in daily incidence to calibrate uncertainty in its probabilistic forecasts. Beating the baseline is typically seen as one marker of a useful model [2] [1], [3]

2.5 Temporal hierarchical forecasting (THieF)

- Add more mathematical detail here - Aggregation: We start with data at a daily scale. Then, based on the aggregation levels of interests for a particular model (e.g. weekly, every two weeks, every four weeks, etc), we induce this sort of artificial aggregation so that we have vectors of data add each of these time scales. Next, forecasts are made using each of these datasets separately. The reason that we are interested in different time scales is that there may be important patterns that are lost add higher time scales amid the noise of lower ones. More often than not, different

patterns will be picked up at different aggregation levels, which means we get forecasts for the same future time periods that may not agree. This is where reconciliation comes into play, in which we combine all of the forecasts that we made.

2.6 Model specifications

2.6.1 THieF models

2.6.2 THieF ensemble models

2.6.3 ARIMA and SARIMA models

2.7 Metrics and Evaluation

2.7.1 Validation phase

2.7.2 Testing phase

CHAPTER 3

RESULTS

3.1 Validation phase

3.2 Testing phase

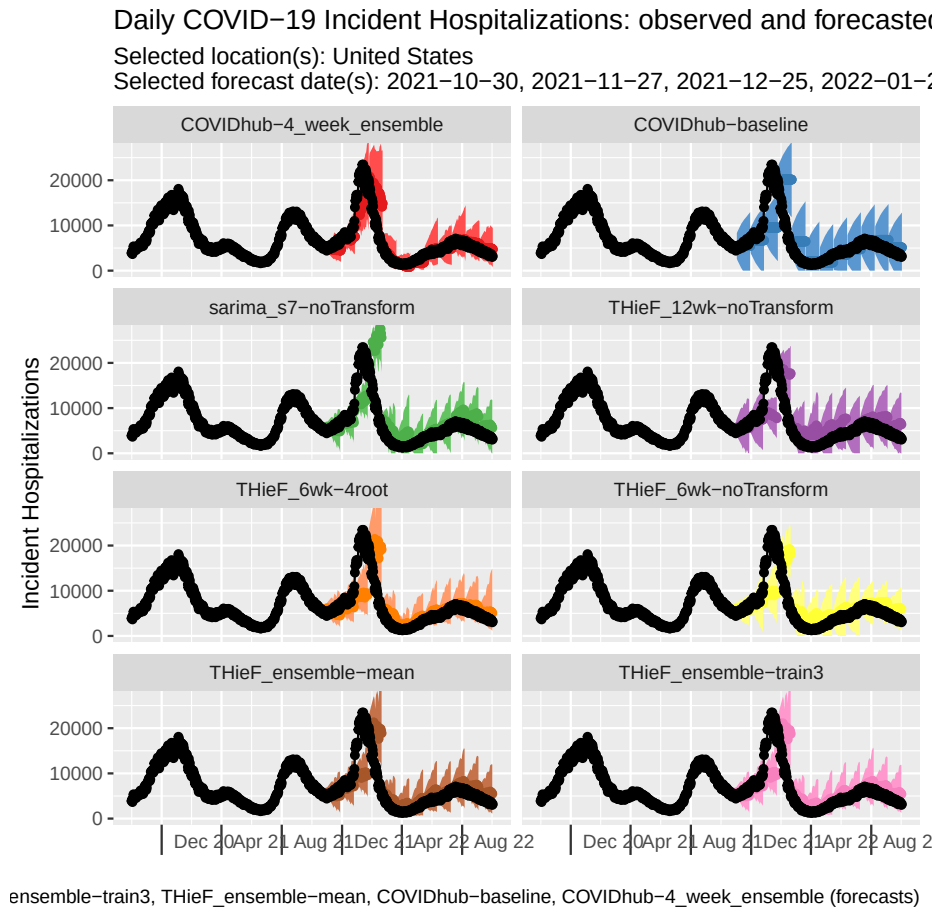


Figure 3.1.

3.2.1 Summary of comparison models

3.2.2 Overall model performance

model	wis	mae	cov_50	cov_95	rwis	rmae
:-----	-----:	-----:	-----:	-----:	-----:	-----:
COVIDhub-4_week_ensemble	1401.3	2085.1	0.629	0.912	0.750	0.829
THieF_6wk-4root	1642.5	2358.8	0.400	0.817	0.880	0.938
THieF_ensemble-train3	1687.4	2443.4	0.358	0.829	0.904	0.972
THieF_ensemble-mean	1700.5	2458.9	0.388	0.836	0.911	0.978
THieF_6wk-noTransform	1727.7	2504.8	0.457	0.862	0.925	0.996
COVIDhub-baseline	1867.5	2514.1	0.715	0.852	1.000	1.000
THieF_12wk-noTransform	2074.1	2908.3	0.367	0.864	1.111	1.157
sarima_s7-noTransform	2316.8	3042.4	0.409	0.798	1.241	1.210

Table 3.1. Summary of overall model performance for the US national level, aggregated across forecast week and 1 to 4 week ahead horizons. Prediction interval (PI) coverage rates measure the fraction of times the fraction of times the 50% and 95% PIs covered the true values. A well-calibrated model should achieve close to nominal (0.5 and 0.95, respectively) coverage rates. The “rwis” and “rmae” show relative weighted interval score and relative mean absolute error, which compare each model to the baseline. The baseline is defined to have relative metrics of 1. Models that perform better than the baseline have relative WIS or MAE less than 1. The table is ordered by ascending WIS value. Thus two THieF-4wk models show the best performance out of the THieF models, meeting the baseline for both relative WIS and MAE.

model	wis	mae	cov_50	cov_95	rwis	rmae
:-----	-----:	-----:	-----:	-----:	-----:	-----:
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model		wis	mae	cov_50	cov_95	rwis	rmae
:-----		-----:	-----:	-----:	-----:	-----:	-----:
COVIDhub-4_week_ensemble		30.181	44.870	0.644	0.945	0.725	0.790
THieF_6wk-4root		34.549	50.116	0.512	0.895	0.830	0.883
THieF_ensemble-train3		35.625	51.574	0.558	0.906	0.856	0.908
THieF_ensemble-mean		35.914	51.805	0.570	0.906	0.863	0.913
THieF_6wk-noTransform		39.508	56.867	0.546	0.878	0.949	1.002
COVIDhub-baseline		41.629	56.772	0.790	0.957	1.000	1.000
THieF_12wk-noTransform		41.656	60.970	0.510	0.877	1.001	1.074
sarima_s7-noTransform		43.538	58.935	0.596	0.868	1.046	1.038

3.2.3 Model performance by horizon week

3.2.4 Model performance by pandemic wave

3.2.5 Model performance by location

3.2.6 Model performance by forecast week

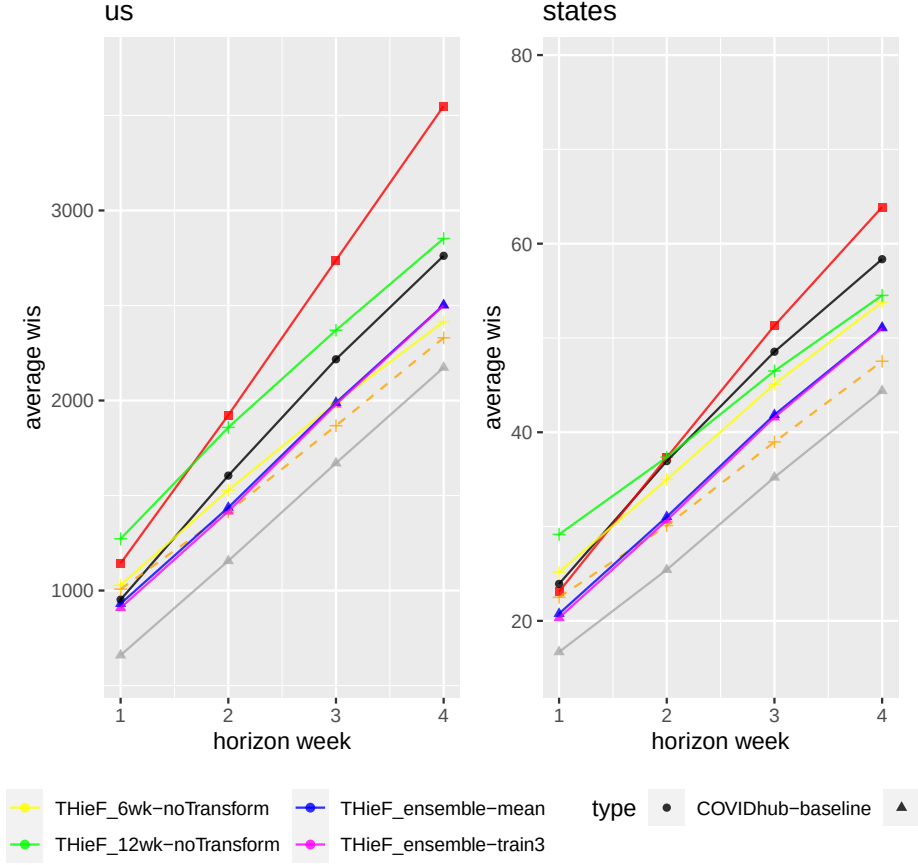


Figure 3.2. Average WIS by the horizon week for each model for the US national level and the states level. The figures show similar trends in WIS when forecasts are stratified by horizon week compared to being aggregated across horizon weeks. All models show worse performance relative to the baseline at the 1-week ahead horizon, likely a result of being overconfident at this horizon. Once again, the THieF-4wk-4root model performs the best for WIS. Plots for average MAE are not shown due to their similarity to the average WIS plots.

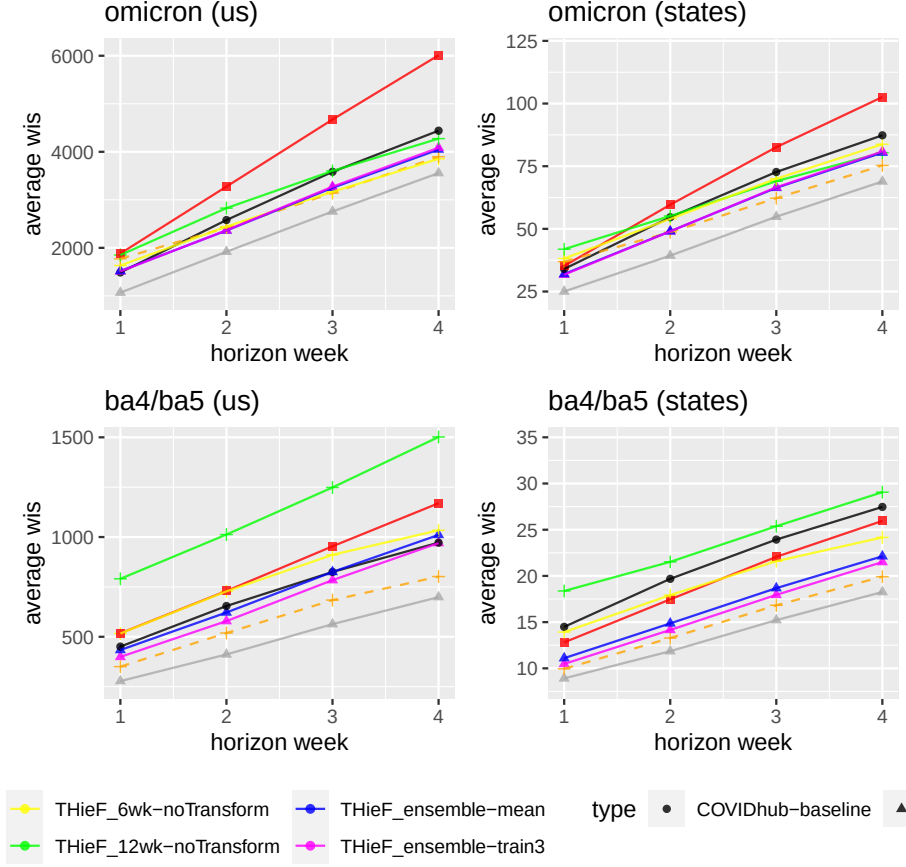


Figure 3.3. Average WIS by the horizon week and pandemic wave for each model for the US national level and the states level. The figures show similar trends in WIS when forecasts are stratified by horizon week and pandemic wave compared to being stratified only across horizon week. Again, all models show worse performance relative to the baseline at the 1-week ahead horizon likely due to being overconfident at this horizon. Model performance is more similar among the top four models during the Alpha wave compared to during the Winter21 wave. Notably, although the THieF-4wk-4root models largely show the lowest WIS, longer horizons during the Winter21 wave saw better performance by the poor-performing THieF-12wk models. Plots for average MAE are not shown due to their similarity to the average WIS plots.

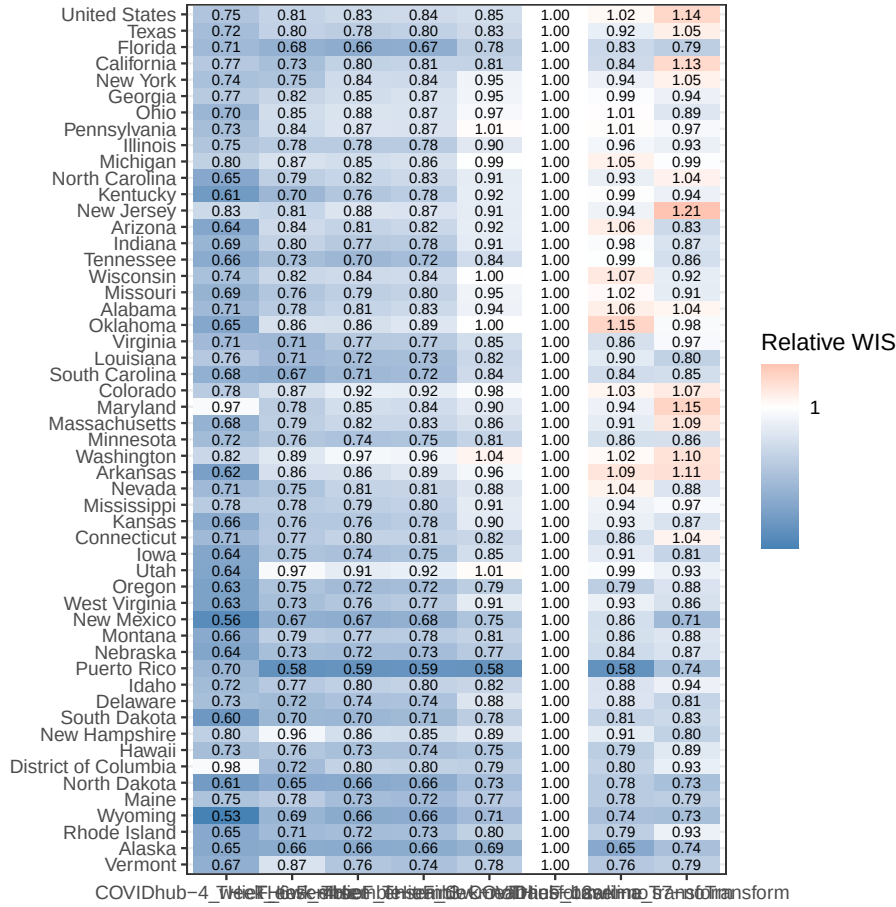


Figure 3.4. Relative WIS by location for each model across all horizons for the entire testing phase. Each square contains the average relative WIS of a model at a particular location, colored based on the relative WIS compare to the baseline. Blue indicates model outperformed the baseline for that location on average, while red indicates the model performed worse than the baseline. Locations are sorted based on cumulative hospitalizations during the testing phase while models are sorted from lowest to highest overall relative WIS, aggregated across states (Table 2). The top three models by relative WIS overall during the testing phase outperformed the baseline in every location.

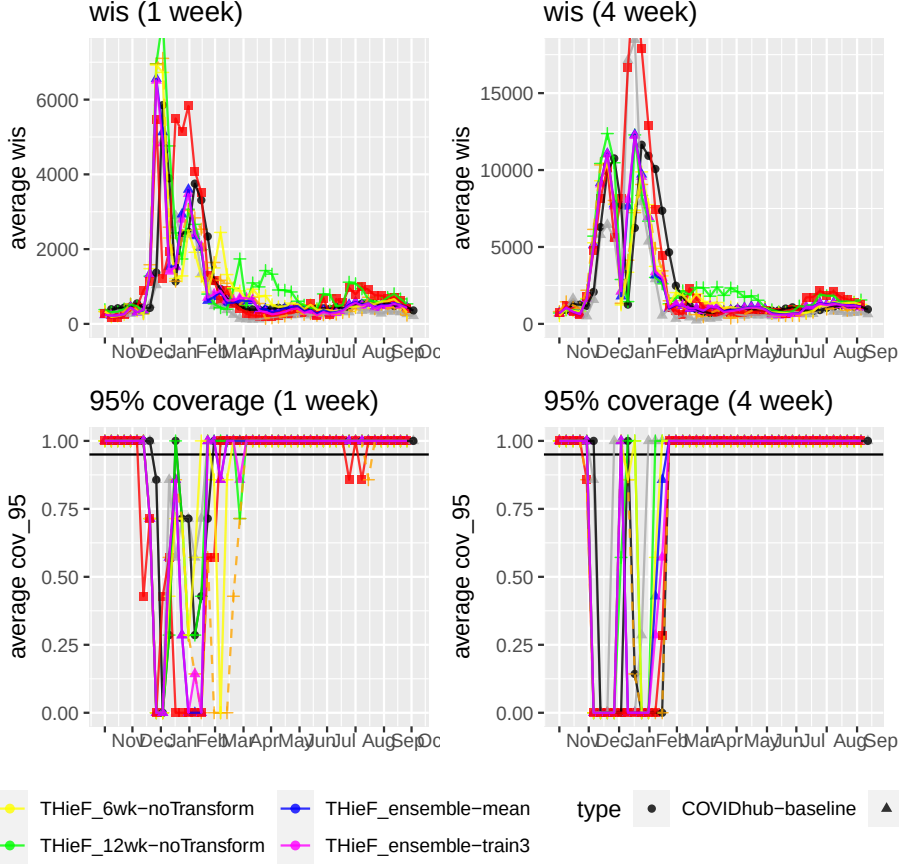


Figure 3.5. Average h-week ahead WIS and 95% PI coverage for each model for the US national level. Model performance differs from week to week: the WIS is the lowest during weeks with stable and low incident hospitalization values. The THieF-4wk models show the best performance with lower or comparable WIS to the baseline during the majority of the training period with especially good performance for 1-week ahead horizons. Meanwhile, the THieF-12wk showed lower or comparable WIS to the baseline during high incidence and changed points, which have been historically found difficult to forecast accurately. Plots for h-week ahead MAE are not shown due to their similarity to the WIS plots. All THieF models show very variable coverage rates, especially for a 1 week ahead horizon, only the models without a data transformation showing stable coverage starting around February 2021.

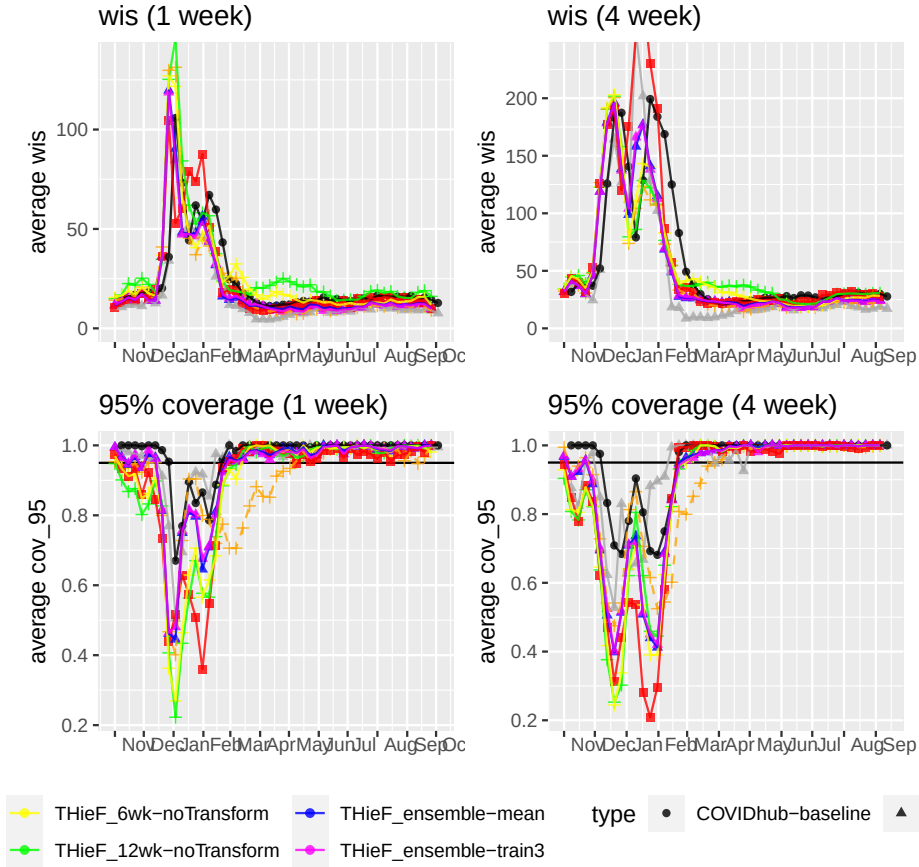


Figure 3.6. Average h-week ahead WIS and 95% PI coverage for each model for the states level. Once again, model performance differs from week to week with the lowest WIS values during weeks with stable and low incident hospitalization values and the THieF-4wk models showing the best performance. However, the THieF-12wk did not show good performance during high incidence and changed points for the states level; instead, THieF-12-4root actually showed some strangely large WIS values during April and May 2021. Plots for h-week ahead MAE are not shown due to their similarity to the WIS plots. One-week ahead coverage rates generally improved as time went on, the best models gaining stable and fairly good coverage rates starting in February 2021. For the 4-week ahead horizon, models without a data transformation showed the best performance and stable coverage starting around February 2021, though THieF-4wk-4root and THieF-8wk-4root also saw decent and fairly stable coverage rates after the downward Winter 21 change point.

CHAPTER 4

DISCUSSION

APPENDIX: SUPPLEMENTAL MATERIALS

model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rm
:-----	-----:	-----:	-----:	-----:	-----:	-----:	-----:
COVIDhub-4_week_ensemble	1	659.8	966.8	0.682	0.937	0.693	0.7
THieF_ensemble-train3	1	909.6	1348.8	0.366	0.859	0.955	1.0
THieF_ensemble-mean	1	930.9	1372.9	0.381	0.859	0.978	1.0
COVIDhub-baseline	1	952.2	1258.8	0.733	0.916	1.000	1.0
THieF_6wk-4root	1	1007.4	1419.1	0.447	0.787	1.058	1.1
THieF_6wk-noTransform	1	1028.7	1467.4	0.465	0.877	1.080	1.1
sarima_s7-noTransform	1	1143.4	1533.9	0.321	0.811	1.201	1.2
THieF_12wk-noTransform	1	1272.4	1793.6	0.402	0.883	1.336	1.4
COVIDhub-4_week_ensemble	2	1156.2	1694.7	0.663	0.920	0.720	0.8
THieF_6wk-4root	2	1413.2	2029.0	0.420	0.847	0.881	0.9
THieF_ensemble-train3	2	1417.1	2046.9	0.393	0.844	0.883	0.9
THieF_ensemble-mean	2	1436.9	2070.8	0.405	0.847	0.895	0.9
THieF_6wk-noTransform	2	1528.5	2177.0	0.469	0.865	0.952	1.0
COVIDhub-baseline	2	1604.9	2113.3	0.742	0.853	1.000	1.0
THieF_12wk-noTransform	2	1858.5	2558.3	0.380	0.853	1.158	1.2
sarima_s7-noTransform	2	1923.1	2536.3	0.408	0.825	1.198	1.2
COVIDhub-4_week_ensemble	3	1670.6	2528.8	0.602	0.900	0.754	0.8
THieF_6wk-4root	3	1867.4	2715.3	0.367	0.840	0.842	0.9
THieF_ensemble-train3	3	1979.6	2845.1	0.351	0.815	0.893	0.9
THieF_6wk-noTransform	3	1989.0	2883.6	0.458	0.853	0.897	0.9
THieF_ensemble-mean	3	1989.0	2860.1	0.389	0.834	0.897	0.9

COVIDhub-baseline		3	2217.0	2993.7	0.712	0.834	1.000	1.0
THieF_12wk-noTransform		3	2369.9	3313.8	0.373	0.865	1.069	1.1
sarima_s7-noTransform		3	2738.4	3581.4	0.458	0.796	1.235	1.1
COVIDhub-4_week_ensemble		4	2173.4	3232.7	0.564	0.891	0.787	0.8
THieF_6wk-4root		4	2330.0	3341.7	0.362	0.795	0.844	0.8
THieF_6wk-noTransform		4	2414.9	3567.4	0.433	0.853	0.875	0.9
THieF_ensemble-train3		4	2501.1	3615.3	0.321	0.795	0.906	0.9
THieF_ensemble-mean		4	2502.2	3613.6	0.375	0.804	0.906	0.9
COVIDhub-baseline		4	2761.4	3782.1	0.670	0.801	1.000	1.0
THieF_12wk-noTransform		4	2852.5	4049.1	0.308	0.856	1.033	1.0
sarima_s7-noTransform		4	3549.4	4630.4	0.452	0.756	1.285	1.2

model		horizon_wk	wis	mae	cov_50	cov_95	rwis	rm
:-----		-----	-----	-----	-----	-----	-----	-----
COVIDhub-4_week_ensemble		1	16.698	25.106	0.646	0.957	0.698	0.7
THieF_ensemble-train3		1	20.324	30.035	0.547	0.927	0.850	0.9
THieF_ensemble-mean		1	20.763	30.521	0.560	0.927	0.868	0.9
THieF_6wk-4root		1	22.500	32.973	0.491	0.892	0.941	1.0
sarima_s7-noTransform		1	23.110	31.921	0.573	0.888	0.966	1.0
COVIDhub-baseline		1	23.921	31.867	0.797	0.970	1.000	1.0
THieF_6wk-noTransform		1	25.168	36.407	0.525	0.893	1.052	1.1
THieF_12wk-noTransform		1	29.174	42.199	0.492	0.886	1.220	1.3
COVIDhub-4_week_ensemble		2	25.421	37.901	0.658	0.951	0.688	0.7
THieF_6wk-4root		2	30.099	43.938	0.515	0.906	0.815	0.8
THieF_ensemble-train3		2	30.663	44.487	0.565	0.913	0.830	0.8
THieF_ensemble-mean		2	31.037	44.798	0.578	0.913	0.840	0.9
THieF_6wk-noTransform		2	34.995	50.267	0.559	0.885	0.947	1.0
COVIDhub-baseline		2	36.947	49.477	0.804	0.960	1.000	1.0

sarima_s7-noTransform		2	37.319	50.281	0.602	0.873	1.010	1.0
THieF_12wk-noTransform		2	37.354	54.313	0.529	0.883	1.011	1.0
COVIDhub-4_week_ensemble		3	35.219	52.519	0.645	0.942	0.726	0.7
THieF_6wk-4root		3	38.976	56.363	0.523	0.899	0.803	0.8
THieF_ensemble-train3		3	41.597	59.884	0.565	0.896	0.857	0.9
THieF_ensemble-mean		3	41.852	60.071	0.577	0.896	0.862	0.9
THieF_6wk-noTransform		3	45.108	64.825	0.555	0.872	0.929	0.9
THieF_12wk-noTransform		3	46.500	68.160	0.519	0.872	0.958	1.0
COVIDhub-baseline		3	48.539	66.299	0.787	0.951	1.000	1.0
sarima_s7-noTransform		3	51.314	69.074	0.605	0.859	1.057	1.0
COVIDhub-4_week_ensemble		4	44.394	65.428	0.626	0.930	0.761	0.8
THieF_6wk-4root		4	47.532	68.482	0.518	0.884	0.815	0.8
THieF_ensemble-train3		4	51.037	73.470	0.554	0.886	0.875	0.9
THieF_ensemble-mean		4	51.111	73.392	0.565	0.885	0.876	0.9
THieF_6wk-noTransform		4	53.803	77.462	0.546	0.861	0.922	0.9
THieF_12wk-noTransform		4	54.518	80.610	0.501	0.866	0.934	0.9
COVIDhub-baseline		4	58.355	81.235	0.773	0.947	1.000	1.0
sarima_s7-noTransform		4	63.886	86.444	0.605	0.849	1.095	1.0
model		horizon_wk	wave		wis	mae	cov_50	cov_95
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COVIDhub-4_week_ensemble		1	ba4_ba5		277.8	378.7	0.890	1.000
THieF_6wk-4root		1	ba4_ba5		350.4	542.9	0.605	0.994
THieF_ensemble-train3		1	ba4_ba5		399.1	685.3	0.459	1.000
THieF_ensemble-mean		1	ba4_ba5		433.6	740.3	0.488	1.000
COVIDhub-baseline		1	ba4_ba5		450.8	486.9	0.924	1.000
THieF_6wk-noTransform		1	ba4_ba5		516.5	822.5	0.599	1.000
sarima_s7-noTransform		1	ba4_ba5		517.8	828.6	0.413	0.988

THieF_12wk-noTransform		1 ba4_ba5		790.7	1319.2	0.349	1.000	1
COVIDhub-4_week_ensemble		2 ba4_ba5		411.4	572.5	0.855	1.000	0
THieF_6wk-4root		2 ba4_ba5		518.8	813.0	0.612	1.000	0
THieF_ensemble-train3		2 ba4_ba5		578.6	966.6	0.491	1.000	0
THieF_ensemble-mean		2 ba4_ba5		621.8	1030.3	0.503	1.000	0
COVIDhub-baseline		2 ba4_ba5		654.2	741.0	0.982	1.000	1
THieF_6wk-noTransform		2 ba4_ba5		727.8	1145.9	0.594	1.000	1
sarima_s7-noTransform		2 ba4_ba5		729.7	1169.6	0.527	1.000	1
THieF_12wk-noTransform		2 ba4_ba5		1012.6	1677.5	0.376	1.000	1
COVIDhub-4_week_ensemble		3 ba4_ba5		563.3	824.7	0.772	1.000	0
THieF_6wk-4root		3 ba4_ba5		683.9	1069.1	0.589	1.000	0
THieF_ensemble-train3		3 ba4_ba5		784.0	1311.5	0.449	1.000	0
COVIDhub-baseline		3 ba4_ba5		825.0	1038.9	1.000	1.000	1
THieF_ensemble-mean		3 ba4_ba5		825.0	1371.3	0.500	1.000	1
THieF_6wk-noTransform		3 ba4_ba5		911.0	1464.4	0.589	1.000	1
sarima_s7-noTransform		3 ba4_ba5		953.5	1514.2	0.570	1.000	1
THieF_12wk-noTransform		3 ba4_ba5		1248.7	2097.1	0.380	1.000	1
COVIDhub-4_week_ensemble		4 ba4_ba5		698.9	1034.3	0.768	1.000	0
THieF_6wk-4root		4 ba4_ba5		801.9	1210.3	0.642	1.000	0
THieF_ensemble-train3		4 ba4_ba5		969.8	1585.1	0.477	1.000	0
COVIDhub-baseline		4 ba4_ba5		972.0	1320.5	1.000	1.000	1
THieF_ensemble-mean		4 ba4_ba5		1010.8	1651.7	0.530	1.000	1
THieF_6wk-noTransform		4 ba4_ba5		1034.0	1631.7	0.609	1.000	1
sarima_s7-noTransform		4 ba4_ba5		1169.6	1821.8	0.603	1.000	1
THieF_12wk-noTransform		4 ba4_ba5		1501.8	2581.8	0.338	1.000	1
THieF_ensemble-mean		1 delta		239.1	305.4	0.857	1.000	0
THieF_ensemble-train3		1 delta		247.9	326.8	0.857	1.000	0

sarima_s7-noTransform		1 delta		270.9	454.3	0.286	1.000	0
THieF_6wk-4root		1 delta		327.6	536.4	0.714	1.000	0
THieF_6wk-noTransform		1 delta		332.5	563.7	0.429	1.000	0
THieF_12wk-noTransform		1 delta		381.8	523.0	1.000	1.000	0
THieF_6wk-noTransform		2 delta		286.5	260.1	1.000	1.000	0
sarima_s7-noTransform		2 delta		290.7	240.0	1.000	1.000	0
THieF_ensemble-mean		2 delta		313.6	215.3	1.000	1.000	0
THieF_ensemble-train3		2 delta		316.2	207.6	1.000	1.000	0
THieF_6wk-4root		2 delta		324.4	280.4	1.000	1.000	0
THieF_12wk-noTransform		2 delta		487.9	404.7	1.000	1.000	0
THieF_6wk-noTransform		3 delta		396.0	511.1	0.857	1.000	0
sarima_s7-noTransform		3 delta		486.3	701.8	1.000	1.000	0
THieF_ensemble-mean		3 delta		509.6	800.2	1.000	1.000	0
THieF_ensemble-train3		3 delta		509.6	797.2	1.000	1.000	0
THieF_6wk-4root		3 delta		547.9	979.9	0.857	1.000	0
THieF_12wk-noTransform		3 delta		585.8	276.3	1.000	1.000	0
THieF_6wk-noTransform		4 delta		530.6	831.2	0.714	1.000	0
THieF_ensemble-mean		4 delta		691.0	1128.0	0.857	1.000	0
THieF_ensemble-train3		4 delta		691.8	1135.1	0.857	1.000	0
THieF_12wk-noTransform		4 delta		710.8	555.6	1.000	1.000	0
sarima_s7-noTransform		4 delta		745.2	1232.6	0.714	1.000	0
THieF_6wk-4root		4 delta		801.6	1496.1	0.429	1.000	0
COVIDhub-4_week_ensemble		1 omicron		1067.9	1595.1	0.460	0.870	0
COVIDhub-baseline		1 omicron		1487.9	2083.4	0.528	0.826	1
THieF_ensemble-train3		1 omicron		1509.9	2136.2	0.240	0.695	3
THieF_ensemble-mean		1 omicron		1517.9	2127.8	0.240	0.695	3
THieF_6wk-noTransform		1 omicron		1632.5	2228.7	0.318	0.734	3

THieF_6wk-4root		1 omicron		1772.1	2437.8	0.260	0.545	3
THieF_12wk-noTransform		1 omicron		1850.9	2381.2	0.435	0.747	4
sarima_s7-noTransform		1 omicron		1881.9	2370.8	0.221	0.604	4
COVIDhub-4_week_ensemble		2 omicron		1919.4	2844.9	0.466	0.839	0
THieF_ensemble-mean		2 omicron		2361.2	3270.0	0.273	0.675	3
THieF_ensemble-train3		2 omicron		2365.4	3287.9	0.260	0.669	3
THieF_6wk-4root		2 omicron		2421.1	3411.4	0.188	0.675	3
THieF_6wk-noTransform		2 omicron		2442.9	3368.8	0.312	0.714	3
COVIDhub-baseline		2 omicron		2579.1	3519.7	0.497	0.702	1
THieF_12wk-noTransform		2 omicron		2827.3	3600.0	0.357	0.688	4
sarima_s7-noTransform		2 omicron		3275.9	4104.9	0.253	0.630	5
COVIDhub-4_week_ensemble		3 omicron		2757.2	4201.2	0.435	0.801	0
THieF_6wk-4root		3 omicron		3141.5	4483.1	0.117	0.669	3
THieF_6wk-noTransform		3 omicron		3167.4	4447.5	0.305	0.695	3
THieF_ensemble-mean		3 omicron		3250.5	4481.1	0.247	0.656	3
THieF_ensemble-train3		3 omicron		3273.1	4511.8	0.221	0.617	3
COVIDhub-baseline		3 omicron		3583.0	4912.1	0.429	0.671	1
THieF_12wk-noTransform		3 omicron		3601.4	4700.2	0.338	0.721	4
sarima_s7-noTransform		3 omicron		4672.1	5833.2	0.318	0.578	5
COVIDhub-4_week_ensemble		4 omicron		3556.3	5294.6	0.373	0.789	0
THieF_6wk-noTransform		4 omicron		3854.4	5589.8	0.247	0.701	3
THieF_6wk-4root		4 omicron		3897.8	5515.5	0.084	0.584	4
THieF_ensemble-mean		4 omicron		4046.9	5650.1	0.201	0.604	4
THieF_ensemble-train3		4 omicron		4084.8	5718.6	0.143	0.584	4
THieF_12wk-noTransform		4 omicron		4274.3	5646.6	0.247	0.708	4
COVIDhub-baseline		4 omicron		4439.6	6090.8	0.360	0.615	1
sarima_s7-noTransform		4 omicron		6010.4	7538.6	0.292	0.506	6

model	horizon_wk	wave	wis	mae	cov_50	cov_95
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COVIDhub-4_week_ensemble	1	ba4_ba5	8.911	13.434	0.719	0.991
THieF_6wk-4root	1	ba4_ba5	9.947	14.828	0.618	0.968
THieF_ensemble-train3	1	ba4_ba5	10.477	15.622	0.674	0.988
THieF_ensemble-mean	1	ba4_ba5	11.101	16.471	0.687	0.990
sarima_s7-noTransform	1	ba4_ba5	12.808	19.000	0.676	0.977
THieF_6wk-noTransform	1	ba4_ba5	13.925	20.539	0.696	0.992
COVIDhub-baseline	1	ba4_ba5	14.488	16.595	0.921	1.000
THieF_12wk-noTransform	1	ba4_ba5	18.372	28.320	0.607	0.989
COVIDhub-4_week_ensemble	2	ba4_ba5	11.854	17.637	0.758	0.996
THieF_6wk-4root	2	ba4_ba5	13.279	19.072	0.687	0.988
THieF_ensemble-train3	2	ba4_ba5	14.153	20.344	0.737	0.994
THieF_ensemble-mean	2	ba4_ba5	14.862	21.204	0.752	0.995
sarima_s7-noTransform	2	ba4_ba5	17.464	25.074	0.751	0.988
THieF_6wk-noTransform	2	ba4_ba5	17.922	25.688	0.767	0.996
COVIDhub-baseline	2	ba4_ba5	19.686	21.668	0.968	1.000
THieF_12wk-noTransform	2	ba4_ba5	21.523	32.823	0.681	0.996
COVIDhub-4_week_ensemble	3	ba4_ba5	15.218	22.860	0.763	0.995
THieF_6wk-4root	3	ba4_ba5	16.821	23.305	0.724	0.993
THieF_ensemble-train3	3	ba4_ba5	17.948	25.285	0.764	0.996
THieF_ensemble-mean	3	ba4_ba5	18.676	26.221	0.777	0.997
THieF_6wk-noTransform	3	ba4_ba5	21.594	31.205	0.779	0.998
sarima_s7-noTransform	3	ba4_ba5	22.064	31.343	0.781	0.992
COVIDhub-baseline	3	ba4_ba5	23.932	27.492	0.975	1.000
THieF_12wk-noTransform	3	ba4_ba5	25.377	39.278	0.697	0.997
COVIDhub-4_week_ensemble	4	ba4_ba5	18.259	27.514	0.757	0.995

THieF_6wk-4root		4 ba4_ba5		19.920	26.570	0.751	0.997
THieF_ensemble-train3		4 ba4_ba5		21.500	29.630	0.776	0.998
THieF_ensemble-mean		4 ba4_ba5		22.126	30.413	0.788	0.999
THieF_6wk-noTransform		4 ba4_ba5		24.166	34.327	0.800	0.999
sarima_s7-noTransform		4 ba4_ba5		25.953	36.718	0.803	0.995
COVIDhub-baseline		4 ba4_ba5		27.465	32.803	0.975	1.000
THieF_12wk-noTransform		4 ba4_ba5		29.061	45.413	0.687	0.999
sarima_s7-noTransform		1 delta		10.381	14.692	0.701	0.975
THieF_ensemble-mean		1 delta		11.595	18.759	0.640	0.995
THieF_ensemble-train3		1 delta		11.872	19.295	0.604	0.995
THieF_6wk-4root		1 delta		14.157	22.679	0.511	0.953
THieF_12wk-noTransform		1 delta		14.886	23.515	0.525	0.948
THieF_6wk-noTransform		1 delta		15.571	25.831	0.511	0.970
sarima_s7-noTransform		2 delta		14.637	20.798	0.695	0.981
THieF_ensemble-mean		2 delta		16.327	26.271	0.654	0.986
THieF_ensemble-train3		2 delta		16.517	26.761	0.640	0.984
THieF_6wk-4root		2 delta		17.992	29.882	0.544	0.978
THieF_12wk-noTransform		2 delta		19.349	30.097	0.596	0.951
THieF_6wk-noTransform		2 delta		19.582	32.199	0.621	0.973
sarima_s7-noTransform		3 delta		23.200	35.709	0.596	0.973
THieF_ensemble-mean		3 delta		25.615	41.494	0.580	0.975
THieF_ensemble-train3		3 delta		25.767	41.935	0.569	0.978
THieF_6wk-4root		3 delta		27.031	44.490	0.519	0.986
THieF_12wk-noTransform		3 delta		27.962	44.220	0.508	0.918
THieF_6wk-noTransform		3 delta		28.402	46.470	0.533	0.945
sarima_s7-noTransform		4 delta		30.502	49.254	0.541	0.945
THieF_ensemble-mean		4 delta		31.711	51.117	0.538	0.967

THieF_ensemble-train3		4 delta		31.797	51.456	0.530	0.970
THieF_6wk-4root		4 delta		32.277	52.556	0.519	0.995
THieF_6wk-noTransform		4 delta		34.183	55.441	0.475	0.929
THieF_12wk-noTransform		4 delta		34.383	54.476	0.489	0.904
COVIDhub-4_week_ensemble		1 omicron		25.018	37.574	0.569	0.921
THieF_ensemble-train3		1 omicron		31.706	46.621	0.403	0.856
THieF_ensemble-mean		1 omicron		31.971	46.748	0.414	0.853
COVIDhub-baseline		1 omicron		33.998	48.183	0.664	0.939
sarima_s7-noTransform		1 omicron		35.196	47.136	0.451	0.784
THieF_6wk-4root		1 omicron		36.899	53.706	0.348	0.804
THieF_6wk-noTransform		1 omicron		38.161	54.611	0.335	0.780
THieF_12wk-noTransform		1 omicron		41.888	58.550	0.362	0.768
COVIDhub-4_week_ensemble		2 omicron		39.326	58.668	0.556	0.906
THieF_6wk-4root		2 omicron		48.671	71.220	0.330	0.815
THieF_ensemble-train3		2 omicron		48.996	71.161	0.377	0.823
THieF_ensemble-mean		2 omicron		49.036	70.919	0.389	0.822
THieF_6wk-noTransform		2 omicron		53.988	77.423	0.334	0.762
COVIDhub-baseline		2 omicron		54.637	77.978	0.635	0.919
THieF_12wk-noTransform		2 omicron		55.135	78.439	0.362	0.759
sarima_s7-noTransform		2 omicron		59.624	78.628	0.439	0.745
COVIDhub-4_week_ensemble		3 omicron		54.848	81.624	0.529	0.891
THieF_6wk-4root		3 omicron		62.249	90.820	0.317	0.798
THieF_ensemble-mean		3 omicron		66.369	95.646	0.372	0.788
THieF_ensemble-train3		3 omicron		66.579	96.197	0.360	0.790
THieF_12wk-noTransform		3 omicron		69.014	98.879	0.336	0.741
THieF_6wk-noTransform		3 omicron		69.993	100.152	0.325	0.740
COVIDhub-baseline		3 omicron		72.688	104.382	0.603	0.903

sarima_s7-noTransform		3 omicron		82.602	109.301	0.425	0.718
COVIDhub-4_week_ensemble		4 omicron		68.905	100.987	0.504	0.869
THieF_6wk-4root		4 omicron		75.300	110.301	0.289	0.767
THieF_12wk-noTransform		4 omicron		80.395	116.309	0.319	0.733
THieF_ensemble-mean		4 omicron		80.412	116.545	0.349	0.770
THieF_ensemble-train3		4 omicron		80.873	117.458	0.337	0.771
THieF_6wk-noTransform		4 omicron		83.755	120.758	0.301	0.723
COVIDhub-baseline		4 omicron		87.327	126.658	0.584	0.897
sarima_s7-noTransform		4 omicron		102.597	136.890	0.414	0.702

BIBLIOGRAPHY

- [1] COVID-19 Forecast Hub. Ensemble model. <https://covid19forecasthub.org/doc/ensemble/>. Accessed: 2023-03-15.
- [2] Cramer, Estee Y., Ray, Evan L., Lopez, Velma K., Bracher, Johannes, Brennen, Andrea, Rivadeneira, Alvaro J. Castro, Gerding, Aaron, Gneiting, Tilmann, Katie H. House, Yuxin Huang, Dasuni Jayawardena, Kanji, Abdul H., Khandelwal, Ayush, Le, Khoa, Mühlemann, Anja, Niemi, Jarad, Shah, Apurv, Stark, Ariane, Wang, Yijin, Wattanachit, Nutch, ..., and Reich, Nicholas G. Evaluation of individual and ensemble probabilistic forecasts of covid-19 mortality in the us. *Proceedings of the National Academy of Sciences* 119, 15 (April 2022), e2113561119.
- [3] Reich, Nicholas G., Tibshirani, Ryan J., Ray, Evan L., and Rosenfeld, Roni. On the predictability of covid-19. <https://forecasters.org/blog/2021/09/28/on-the-predictability-of-covid-19/>, September 2021. Accessed: 2022-04-06.